Operating System Project

‘CPU Scheduling’

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The OUTPUT:

1st Algorithm : FCFS

```
run:
First-Come-First-Serve (FCFS) Scheduling:
Average Waiting Time = 34.125
Average Turnaround Time = 45.375
CPU Finish Time = 90
Total Context Switches = 8
```

2nd Algorithm : RR

When the q=200

```
Round Robin (RR) Scheduling (Time Quantum = 200ms):
Average Waiting Time = 40.0
Average Turnaround Time = 45.375
CPU Finish Time = 90
Total Context Switches = 8
BUILD SUCCESSFUL (total time: 0 seconds)
```

When the q = 20

```
Round Robin (RR) Scheduling (Time Quantum = 20ms):
Average Waiting Time = 36.875
Average Turnaround Time = 41.625
CPU Finish Time = 85
Total Context Switches = 8
BUILD SUCCESSFUL (total time: 0 seconds)
```

Code Analysis:

From the Output above we concluded that Round Robin algorithm with both time quantum provides a lower waiting time and turnaround time compared to First-Come-First-Serve algorithm. To be more detailed the Robin Round algorithm when the q=200 preform better when the q=20, and it acts similar to First-Come-First-Serve algorithm which means it provide low waiting time and turnaround time for all the processes.

The difference between the two Round Robin algorithms: when the q= 20 the algorithm preforms context switching more repeatedly which result a higher waiting time and turnaround time for some processes.

In the other hand, when the q=200 as we mentioned before it acts similar to the first-come-first-serve algorithm which result a lower waiting time and turnaround time for all processes.